

A Cost-Effective Arduino-Based System for Measuring Solar Panel Stability

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Abstract

Perovskite solar cells have achieved high power conversion efficiencies, but their long-term stability requires improvement for widespread commercial adoption. Efficient stability testing is critical for advancing this technology. However, the high cost of commercial stability analyzers (\$10,000–\$60,000) presents a significant barrier for many research laboratories.

We have developed an open-source Stability Measurement System (SMS) based on the Arduino platform. This system provides essential measurement capabilities at approximately 1% of the cost of commercial alternatives. Each \$100 SMS module enables simultaneous current-voltage (JV) characterization and maximum power point tracking (MPPT) on eight separate solar cells. The system’s modular design allows for expansion to monitor over 250 cells concurrently. The SMS consists of custom printed circuit boards, 3D-printed sample holders, and Python-based control software. Minimal skills are required to assemble.

Validation against a commercial system demonstrated acceptable measurement accuracy, with relative standard deviations of 1.26% for fill factor and 0.34% for power conversion efficiency. A 200-hour stability test confirmed the system’s reliability for long-term studies. While variability in contact resistance limits the system’s precision for absolute benchmarking, the SMS is effective as a screening platform for comparative stability studies. By providing greater access to high-throughput aging studies, this open-source platform can facilitate broader exploration of perovskite compositions and accelerate research toward commercial viability.

1 Introduction

Perovskite solar cells (PSCs) have advanced rapidly, with power conversion efficiencies (PCEs) increasing from approximately 4% in 2009 to over 25.5% within a decade. They are now rivaling the performance of silicon-based cells while offering simpler and more cost-effective manufacturing processes. [?, ?] However, their limited operational stability remains a significant obstacle to commercialization. Environmental factors such as illumination, moisture, oxygen, thermal stress, and electrical bias can induce ion migration, interfacial reactions, and phase segregation, all of which contribute to substrate degradation. [?, ?, ?, ?, ?]

To address these stability challenges, specialized instrumentation is required to apply stress conditions and perform electrical measurements. However, commercial stability analyzers are expensive, with costs ranging from \$10,000 to \$60,000. [?] Lower-cost commercial systems exist, but are often limited to single-cell measurements. [?] These high costs restrict the ability of many research groups to conduct comprehensive studies on combinatorial compositional spaces or to obtain statistically significant data from large substrate populations.

In recent years, the open-source hardware movement has provided a practical approach to making high-throughput stability testing more accessible. Community-driven initiatives have produced affordable environmental chambers, LED-based solar simulators, and optical monitoring systems. [?, ?, ?] Although some low-cost, open-source systems for single-channel current-voltage (JV) and maximum power point tracking (MPPT) have been developed, they often require specialized knowledge for reproduction and use. [?, ?, ?] Currently, there is no open-source instrument capable of performing automated, multi-

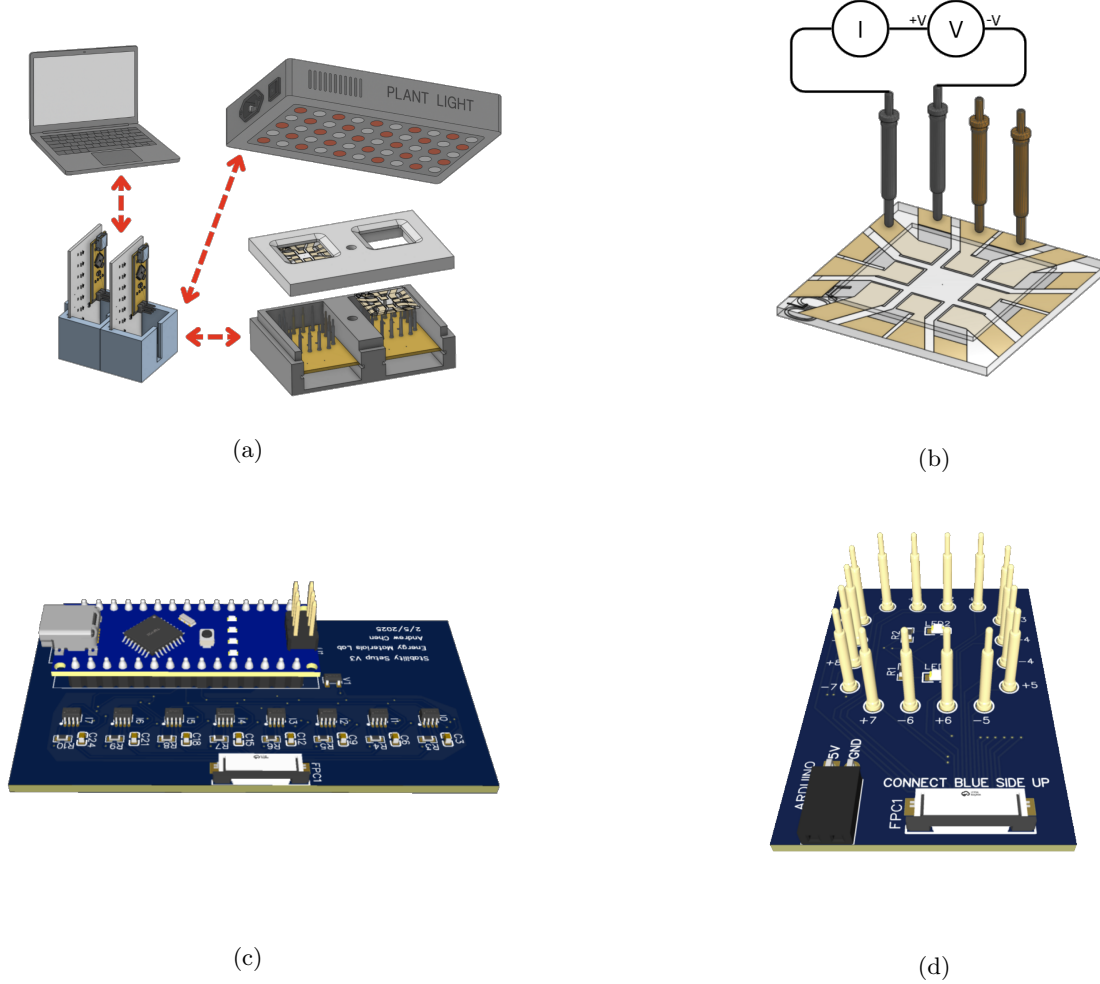


Figure 1: Physical overview of the Stability Measurement System (SMS). a) Overall system block diagram. Red arrows indicate data and power connections. b) Detail of the pogo-pin contacts in the 3D-printed enclosure. c) Main PCB assembly: eight independent, electrically isolated measurement channels. d) Sample holder mating with eight cells via spring-loaded pogo pins.

cell parallel MPPT and JV measurements.

This work introduces a \$100 Arduino-based Stability Measurement System (SMS) designed to fill this gap. Each SMS module includes a custom printed circuit board (PCB), a sample holder, a 3D-printed enclosure, and Python-based control software/graphical user interface (GUI) to perform parallel JV and MPPT measurements on eight independent pixels. Multiple modules can be connected to a single Windows computer, enabling the supervision of over 250 cells in parallel. The system requires only basic soldering, 3D printing, and computer skills for assembly.

The SMS provides the core functionality of commercial analyzers at approximately 1% of the cost. We benchmark the SMS against a commercial system, demonstrate its reliable operation over a 200-

hour period, and release all hardware and software under an open-source license to encourage community adoption and further development [?].

2 System Description

At the core of each SMS module is a two-layer printed circuit board (PCB) (Fig. ????) that includes:

- An Arduino microcontroller for sequencing and USB communication.
- Eight 12-bit Digital-to-Analog Converters (DACs) that supply a programmable bias voltage (0–3.3 V, 0.8 mV steps).

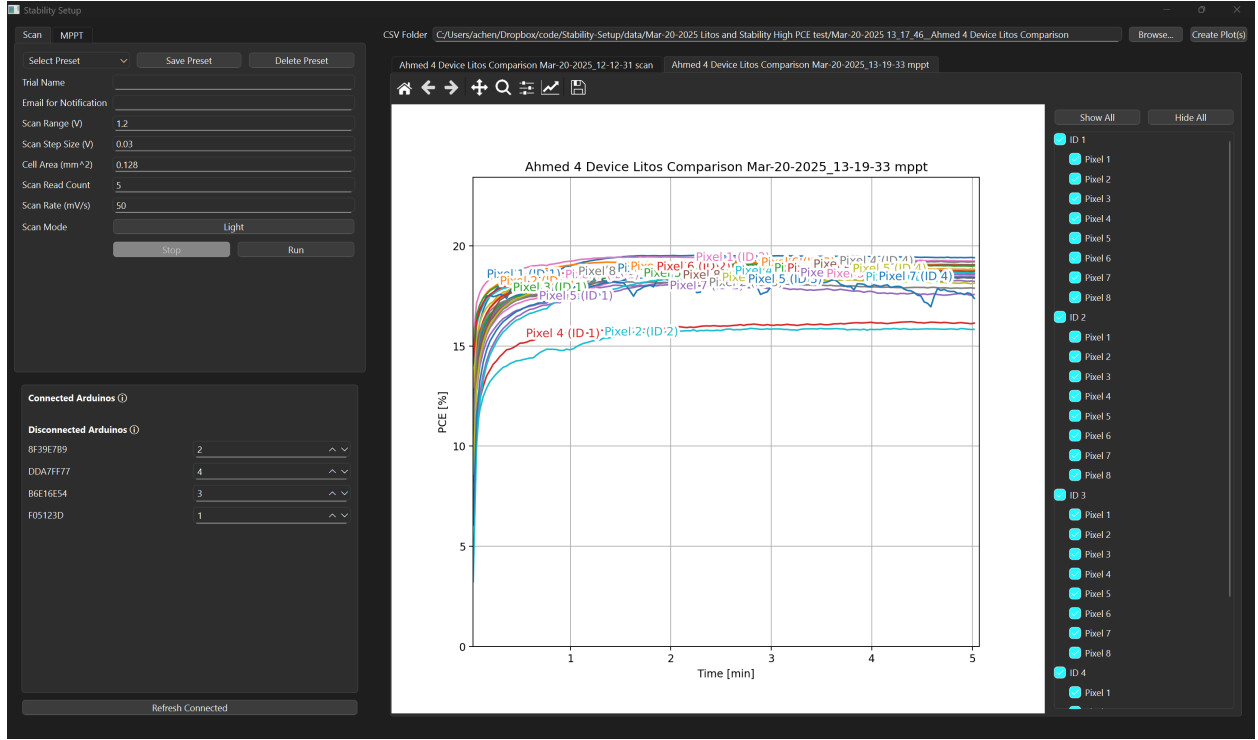


Figure 2: Screenshot of the SMS GUI running in Windows 11, displaying the results of a 32-cell, 5-minute MPPT trial.

- Eight dual-channel Analog-to-Digital Converters (ADCs) for measuring voltage (-0.3 – 26 V, 4 mV resolution) and current (0 – 32 mA, 8 μ A resolution).

Each of the eight channels is fully isolated, allowing for simultaneous JV sweeps or MPPT on all cells. Each channel operates independently with its own dedicated DAC output and ADC input, which prevents electrical crosstalk between measurements. This isolation ensures that the electrical characteristics of one cell do not affect the measurements of adjacent channels, enabling true parallel testing.

The cell holder is a companion PCB that connects to the main board via a 16-pin ribbon cable, with two conductors per channel. Two additional breadboard wires power built-in LEDs for mask alignment. Each cell is contacted by spring-loaded pogo pins within a 3D-printed frame (Fig. ???), which ensures consistent pressure on both electrodes. With a 600 mm ribbon cable, the resistance of each channel is $2\ \Omega$ between the pogo pin tip and the SMS input/output. The 3D-printed frame also includes a press-fit port for a 2.5 mm internal diameter N_2 line. Illumination is provided by a commercially available LED grow light.

Power and data are transmitted over a single USB

cable to a host PC. System capacity can be expanded by connecting additional SMS modules via USB. A powered USB hub is recommended when using more than four modules per computer USB port. We have verified stable parallel data collection from eight SMS systems (64 cells) on a Windows 10/11 desktop. A minimum of 12 GB of RAM is recommended, and any modern CPU should be sufficient. The theoretical limit is estimated to be over 32 SMS systems operating in parallel (256 cells).

2.1 Software

The SMS is controlled by a Python-based GUI that runs on Windows 10/11. The software allows users to configure and run JV and MPPT measurements, view real-time data, and log results to CSV files. The GUI is designed to be intuitive and user-friendly, and a comprehensive user guide is available in the open-source repository.

Figure ?? shows the Python/PyQt GUI. Its key features include:

- Interactive configuration of scan parameters (e.g., voltage range, step size, sampling rate).
- A test queue for automated saving of test sequences.

- Real-time plotting of JV curves and MPPT power traces.
- Continuous CSV logging for robust data archiving, which protects against data loss in the event of a PC crash.

2.2 JV Scan

To characterize initial performance and detect hysteresis, the system conducts a bidirectional voltage sweep (0–3.3 V). At each step, it averages multiple readings to reduce noise. The system then computes the open-circuit voltage (V_{OC}) and maximum power point voltage (V_{MPP}). The configurable parameters are as follows:

1. **Scan Range:** The voltage span for the scan (0–3.3 V).
2. **Scan Step Size:** The voltage increment between readings (0.001–1 V).
3. **Cell Area:** The active area of the solar cell (cm^2), used to calculate current density.
4. **Scan Read Count:** The number of readings averaged per voltage step to reduce noise.
5. **Scan Rate:** The speed of the voltage sweep (mV/s).
6. **Measurement Delay:** The delay between setting the voltage and the measurement, which allows the cell to stabilize (ms).
7. **Measurement Interval:** The time interval across which measurements are taken per voltage step (ms).
8. **Measurement Time:** The total duration for the MPPT process (specified in minutes or hours).

The P&O method was selected for its straightforward implementation and low computational demand, which is well-suited for Arduino-based systems. In contrast to methods such as incremental conductance or fuzzy logic control, P&O does not require extensive calibration or complex algorithms, which improves its reliability. Since our experiments are conducted under stable laboratory conditions, the risk of common P&O issues, such as oscillation around the MPP or convergence to local minima, is minimal. This makes P&O a practical and reliable choice for our testing setup [?].

2.4 Cost and Assembly

The majority of the SMS system can be factory-fabricated and assembled, with a minimum order quantity of five per board, for a total cost of under \$275 (excluding shipping and duties) [?]. Complete assembly instructions and design files are provided in the open-source repository [?].

2.3 MPPT Algorithm

We implemented a perturb-and-observe (P&O) tracking algorithm [?]. The initial voltage can be set manually or automatically by using a prior JV scan to set the start voltage at 85% of V_{OC} [?]. The configurable parameters are as follows:

1. **Starting Voltage:** The initial load voltage (0–3.3 V) (does not apply if a scan is conducted beforehand).
2. **Starting Percentage:** The percentage of V_{OC} to set the starting voltage (only applies if a scan is conducted beforehand).
3. **Step Size:** The voltage increment for adjustments (0.001–1 V).
4. **Cell Area:** The active area of the solar cell (in cm^2) for live PCE calculations.
5. **Measurements Per Step:** The number of readings per voltage level that are averaged for noise reduction.

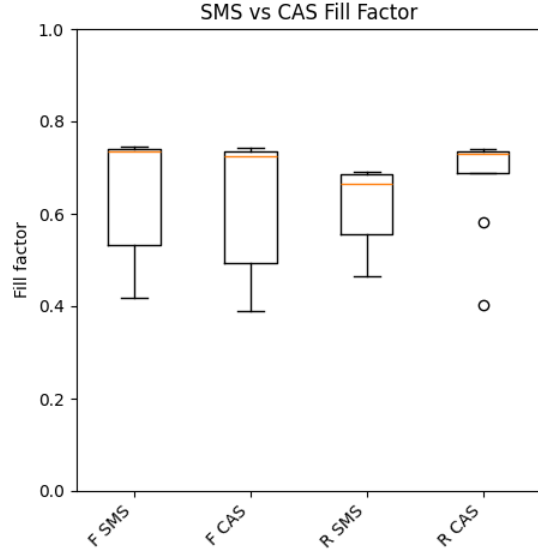
3 Experimental

3.1 Instruments and Calibration

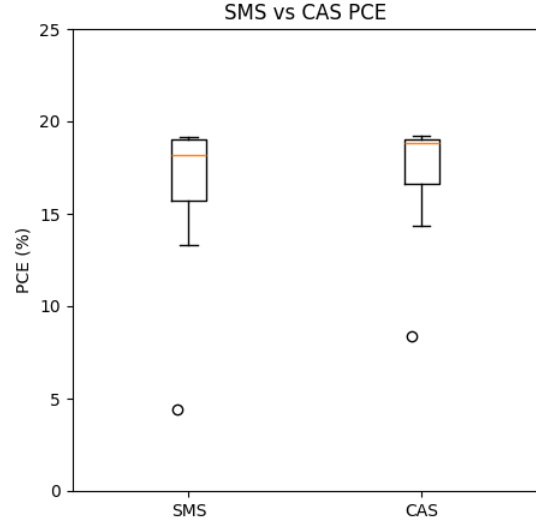
A WaveLabs Sinus 70 solar simulator [?], calibrated to 1 sun (100 mW/cm^2) with an AM 1.5G spectrum, was used for all JV and MPPT comparison measurements between the SMS and the commercial analysis system (CAS). For stability testing, a cost-effective LED grow light was adjusted to deliver 1 sun intensity, as validated by a calibrated reference solar panel.

3.2 substrate Fabrication

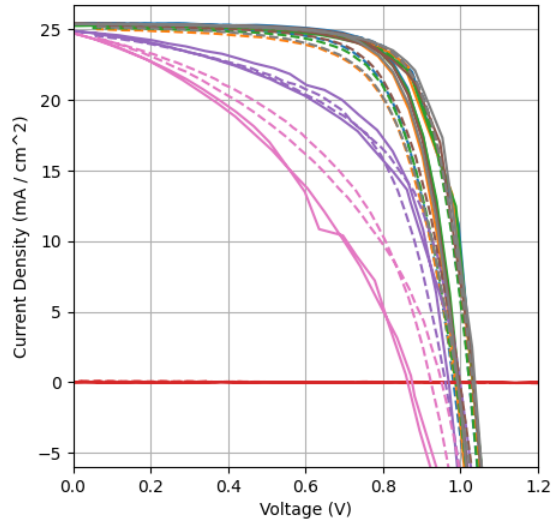
Reference perovskite solar cells (PSCs) were fabricated with eight cells per substrate (32 cells total). The substrates followed the n-i-p stack architecture reported by Kim et al. [?]: FTO / compact TiO_2 / mesoporous TiO_2 / PEAi interlayer / $\text{Cs}_{0.09}\text{FA}_{0.91}\text{PbI}_3$ perovskite / PEAi interlayer / Li-TFSI, tBP-doped spiro-OMeTAD / Au. Each cell



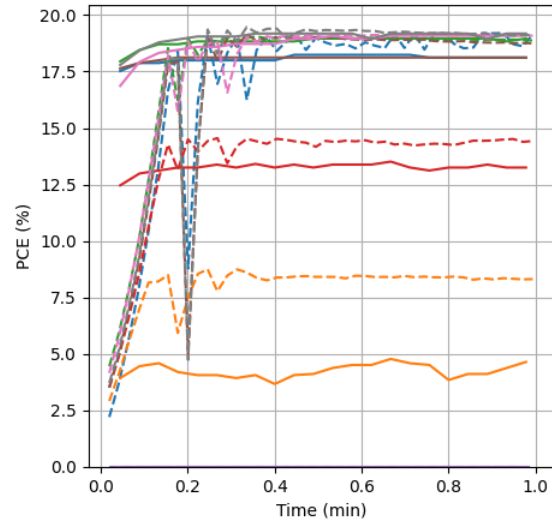
(a)



(b)



(c)



(d)

Figure 3: (a) Box plot of SMS vs CAS Fill Factor on all 4 substrates, with forward and reverse JV scans separated. (b) Box plot of SMS vs CAS PCE on all 4 substrates (c) JV scan of one substrate from SMS plotted against CAS. Dotted traces represent SMS while solid traces represent CAS. (d) MPPT plot of one substrate from SMS plotted against CAS. Dotted traces represent SMS while solid traces represent CAS.

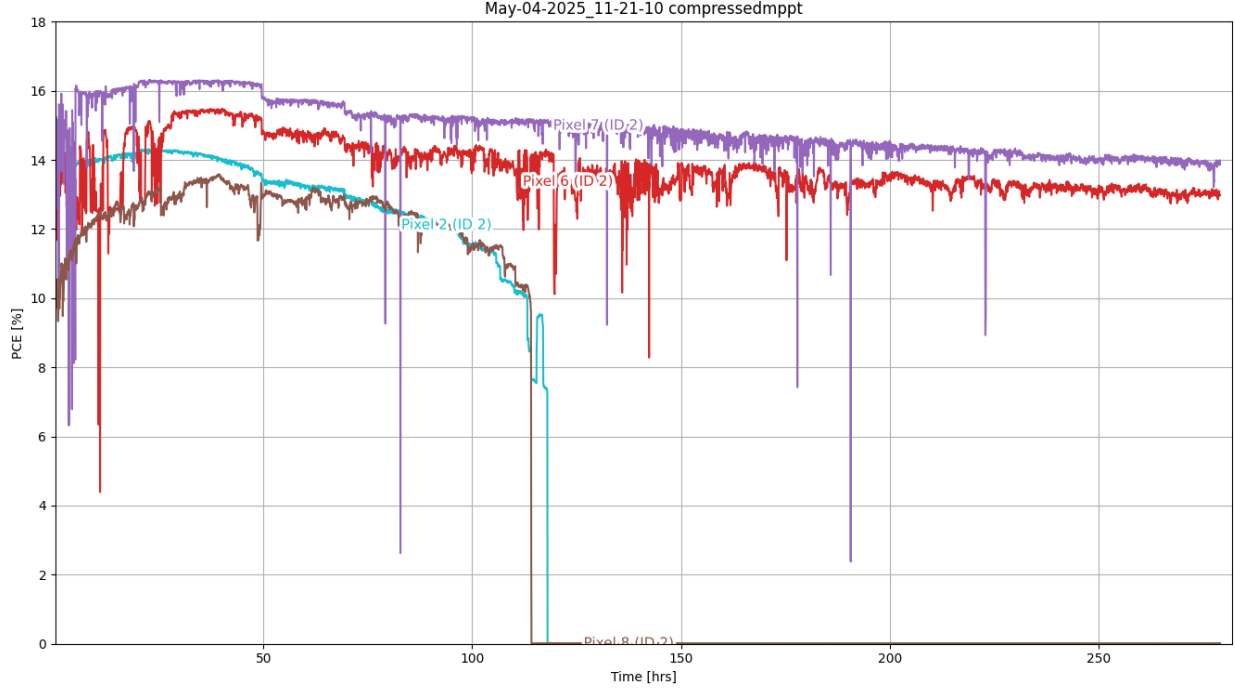


Figure 4: 200-hour stability measurement.

had an active area of approximately 0.128 cm^2 . To ensure consistent electrical contact during long-term testing, silver paste was applied to the cell contacts. The use of silver paste, while ensuring stable long-term contact, introduces a potential source of variability in contact resistance compared to methods like evaporated metal contacts, which may affect the absolute accuracy of measurements.

3.3 Metrics

The primary metrics of interest were the fill factor (FF), open-circuit voltage (V_{OC}), short-circuit current density (J_{SC}), and maximum power conversion efficiency (PCE). The FF, J_{SC} , and V_{OC} were determined from the JV scans, while the PCE was calculated from the stabilized output of the MPPT traces.

3.4 Silicon-Diode Baseline Protocol

A silicon diode was measured to validate the baseline accuracy of the SMS. The diode was placed in both the SMS and CAS holders and measured by their respective systems. JV scans were conducted from 0 V to 1.2 V at a rate of 50 mV/s and the results were compared.

3.5 Separate Holder Protocol

Four PSC substrates were measured sequentially using the SMS and CAS in their respective holders. The substrates were first placed in the SMS holder and then transferred to the CAS holder for measurement, with a waiting period of less than 10 minutes between measurements. Nitrogen purging was used in both trials. No optical masks were used due to the potential for misalignment when transferring samples between holders. The substrates were exposed to oxygen for short periods during handling and transfer.

Each measurement consisted of a forward JV sweep from 0 V to 1.2 V, a reverse JV sweep back to 0 V at 50 mV/s, and a 1-minute MPPT trial. The SMS voltage step size was fixed at 0.03 V, while the CAS step sizes varied internally. The MPPT trial was initiated at 85% of the V_{OC} determined from the preceding JV scan. The voltage step size for the MPPT was 0.005 V on the SMS, while the CAS step sizes varied internally. PCE was calculated from the average of the final 30 s of each trace.

3.6 Same Holder Protocol

To eliminate errors associated with substrate handling and exposure, an adapter board was used to connect the SMS directly to the CAS sample holder. Four PSC substrates were tested using this setup.

Table 1: Agreement statistics for SMS vs. CAS (Each system with its respective holder).

Metric	Mean Difference	MAE	RMSE	95% CI	95% Limits of Agreement
Fill factor (%)	-0.55	6.49	11.89	(-3.54, +2.44)	(-24.01, +22.90)
V_{OC} (V)	+0.0244	0.0916	0.2307	(-0.0334, +0.0821)	(-0.4288, +0.4775)
J_{SC} (mA cm⁻²)	+1.3222	1.9173	2.3526	(+0.8323, +1.8121)	(-2.5218, +5.1663)
PCE (%)	+0.4628	1.9145	2.6241	(-0.5572, +1.4827)	(-4.6928, +5.6184)

Table 2: Agreement statistics for SMS vs. CAS (CAS holder on both systems).

Metric	Mean Difference	MAE	RMSE	95% CI	95% Limits of Agreement
Fill factor (%)	-2.21	5.29	6.94	(-3.89, -0.52)	(-15.22, +10.81)
V_{OC} (V)	+0.0049	0.0158	0.0283	(-0.0022, +0.0121)	(-0.0502, +0.0600)
J_{SC} (mA cm⁻²)	-0.3306	0.4229	0.7264	(-0.4962, -0.1650)	(-1.6086, +0.9474)
PCE (%)	-0.4478	0.9822	1.6322	(-1.0439, +0.1483)	(-3.5767, +2.6811)

Table 3: Platform repeatability statistics for fill factor and power.

Metric	Fill factor (units)	Power (mW, MPPT derived)
Repeatability σ_r	0.0087	0.0331
RSD (%)	1.26%	0.34%
95% Repeatability Coefficient	0.0241	0.0918

Masks were used in this experiment, reducing the active area to 0.0625 cm². The JV and MPPT protocols were the same as in the separate holder experiment.

3.7 Repeatability Protocol

To quantify the precision of the measurement system, we recorded four complete JV scans and MPPT measurements on four PSC substrates. The relative standard deviation (RSD) was calculated for the FF from the JV data and for the stabilized PCE from the MPPT data.

3.8 200-Hour Stability Protocol

A 200-hour degradation test was performed on 32 parallel solar cells under illumination from the LED grow light. While the spectrum of the LED grow light does not fully replicate the AM 1.5G solar spectrum, it provided adequate conditions for a controlled degradation and repeatable stability assessment under laboratory conditions. A moving average was applied to the long-term stability data to reduce noise and clarify degradation trends.

4 Results and Discussion

4.1 Diode Baseline Results

The diode JV scans showed absolute differences between the SMS and the CAS of 0.035 for FF, 0.128 mA for J_{SC}, and 8 mV for V_{OC}, which correspond to relative errors of 6.21%, 5.7%, and 0.16% respectively.

4.2 Separate Holder Results

The JV characterization results for 32 PSC cells are summarized in Table ??, and Figure ???? illustrates the discrepancies in fill factor. The mean difference was minimal for fill factor (FF) at -0.0055, moderate and statistically insignificant for open-circuit voltage (V_{OC}) at +24.4 mV, and significant for short-circuit current density (J_{SC}) at +1.32 mA cm⁻². The broad limits of agreement suggest considerable random variability, which is likely due to inconsistent contact resistance and illumination uniformity. These results indicate that the SMS is best suited for rapid screening rather than for precision benchmarking.

MPPT trials showed an average power conversion efficiency (PCE) offset of +0.46% (statistically insignificant), with substantial variability (limits of agreement of $\pm 5.16\%$).

4.3 Same Holder Results

Using the CAS holder for both systems significantly reduced the variability associated with sample transfer and handling. The JV and PCE metrics showed improved agreement (Table ??), with smaller mean differences and narrower limits of agreement. This demonstrates that a significant portion of the scatter observed in the separate holder experiments originated from the custom sample holder rather than the measurement board. Variations in the spring-loaded pogo pin pressure between the custom-designed holder and the commercial one were likely the largest contributor to this error. The commercial holder, designed for high-precision measurements, likely provides more uniform contact pressure and better sample alignment than our 3D-printed version.

4.4 Repeatability Analysis

To evaluate the intrinsic precision of the SMS, we performed four consecutive JV and MPPT measurements on a single PSC substrate. As summarized in Table ??, the relative standard deviations (RSD) were 1.26% for fill factor (FF) and 0.34% for maximum power conversion efficiency (PCE). These low RSD values indicate that the SMS provides highly consistent measurements when substrate handling is minimized.

4.5 Long-term Stability Analysis

The 200-hour degradation test under LED illumination demonstrated the SMS’s suitability for reliable long-term monitoring. The use of silver paste contacts facilitated stable measurements over the duration of the test. The observed degradation trends were clear and reproducible, which validates the system’s utility for stability assessments under controlled laboratory conditions.

5 Future Work

Several enhancements could further improve the SMS’s measurement accuracy and expand its capabilities for comprehensive stability testing:

5.1 Environmental Control Integration

Integrating the SMS with open-source temperature and humidity chambers [?] would allow for the systematic investigation of thermal and moisture stress

effects on perovskite solar cells. This would create a complete environmental testing platform, while maintaining a cost-effective approach, and would enable researchers to study degradation mechanisms under realistic operating conditions, including temperature cycling and humidity exposure.

5.2 Four-Wire Kelvin Sensing

Implementing four-wire Kelvin sensing would lessen the contact resistance errors that currently contribute to measurement variability. This would involve separating the current-carrying and voltage-sensing paths, which would provide more accurate voltage measurements, particularly for low-resistance substrates. The improved precision would make the SMS more suitable for absolute benchmarking applications.

5.3 Enhanced Sample Holder Design

A more robust sample holder with an improved mechanical design could reduce the contact pressure variability observed in our experiments. Potential improvements include: (1) a uniform spring-loaded contact mechanism with calibrated force application, (2) better alignment features to ensure reproducible cell positioning, and (3) more reliable N₂ purging system. These modifications would significantly reduce the measurement scatter observed when transferring samples between different holders.

5.4 Higher-Precision Sensing Components

Upgrading to more accurate current and voltage sensing integrated circuits would improve measurement resolution and reduce systematic errors. Higher-resolution ADCs (16-bit or greater) and precision current sensors with lower offset drift would enhance the system’s ability to detect subtle changes in substrate performance during long-term stability studies. This would be particularly beneficial for tracking gradual degradation trends that might be masked by the current levels of measurement noise.

6 Conclusion

We have developed and validated an Arduino-based Stability Measurement System (SMS) to address the need for accessible, high-throughput perovskite solar cell stability testing. At a cost of approximately \$100 per 8-channel module, the SMS provides core measurement functionality at about 1%

of the cost of commercial analyzers, which range from \$10,000 to \$60,000. The SMS enables parallel testing of over 250 cells, overcoming the single-channel limitation of many existing low-cost systems.

Validation against a commercial system demonstrated acceptable measurement accuracy. The SMS achieved relative standard deviations of 1.26% for fill factor and 0.34% for power conversion efficiency under controlled conditions. Although the SMS has reduced accuracy and precision compared to commercial systems, it is well-suited for its intended application as a cost-effective screening platform for comparative stability studies. A 200-hour degradation test confirmed the SMS’s reliability for extended monitoring under laboratory conditions.

The open-source design, which requires only basic 3D printing and computer skills for assembly, improves access to multi-channel stability testing. This allows research groups with limited budgets to explore combinatorial compositional spaces and acquire statistically significant data on substrate populations—capabilities previously limited by the high cost of commercial instrumentation.

By reducing the financial barriers to high-throughput aging studies, the SMS can help accelerate research and development in perovskite solar technologies, bringing the field closer to large-scale commercial viability.

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